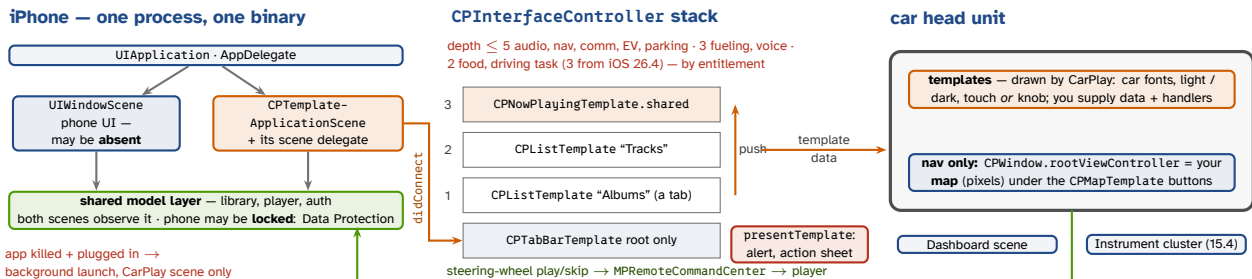


In one line: A CarPlay app is **not a separate target**: it is a second **scene** (CPTemplateApplicationScene) of your iOS app, same process. You never draw its UI — you hand **templates** to a CPInterfaceController and the **car renders them**. Apple grants one **category entitlement**; the category decides which templates you may use and how **deep** the stack goes. Only **navigation** apps draw pixels (the map, in a CPWindow).



How it works

- **Entitlement** per category, requested from Apple; a Boolean key in `.entitlements`, matching App ID + profile. A template outside your category → **runtime exception**.
- **Scene manifest:** `UISceneConfigurations` gets a `CPTemplateApplicationSceneSessionRoleApplication` entry (class `CPTemplateApplicationScene` + your delegate). No extension, no new target.
- **Connect:** `templateApplicationScene(_:didConnect:)` hands you the `CPInterfaceController`: keep it, `setRootTemplate`. Nav apps get `...didConnect:to:` with a `CPWindow`. Disconnect → drop both.
- **Stack:** push / pop / pop(to:); `presentTemplate` shows one modal. `CPTabBarTemplate` is root only, each tab its own stack.
- **Templates are data:** text, `CPImageSet` (light/dark), handlers. Update in place (`updateSections(_:)`).
- **The car sets limits:** `maximumItemCount`, `maximumTabCount`, `CPSessionConfiguration.limitedUserInterfaces` (`.keyboard`, `.lists while driving`); a grid shows 8 buttons.

Categories — key `com.apple.developer.carplay-`

category	key suffix	depth	own templates · rule
audio	audio	5	Now Playing
communication	communication	5	Contact
navigation	maps	5	Map, Search, Voice
EV charging, parking	charging, parking	5	Point of Interest, Information
quick food ordering	quick-ordering	2 (3 [†])	Point of Interest, Information
fueling (iOS 16)	—	3	
driving task (iOS 16)	driving-task	2 (3 [†])	
public safety	—	—	dispatch, routing
voice conv. (26.4)	voice-based-conversation (unverified)	3	audio session only while voice is in use
video (iOS 27)	—	—	parked only, AirPlay video

[†] iOS 26.4+. List, grid, tab, alert, action sheet: all. Information: not audio. EV + fueling combine, audio + video too. Old `MPPlayableContentManager`: deprecated (14).

Example — audio app, scene delegate

```
func templateApplicationScene(_ s: CPTemplateApplicationScene,
                             didConnect ui: CPInterfaceController) {
    self.ui = ui // keep it
    let max = CPLISTTemplate.maximumItemCount // the car's limit
    let rows = Library.shared.albums.prefix(max).map { a in
        let row = CPLISTItem(text: a.title, detailText: a.artist)
        row.handler = { _, done in
            Player.shared.play(a) // shared model
            ui.pushTemplate(CPNowPlayingTemplate.shared,
                           animated: true, completion: nil)
            done() // or it spins forever
        }
        return row
    }
    let list = CPLISTTemplate(title: "Albums",
                             sections: [CPLISTSection(items: rows)])
    ui.setRootTemplate(list, animated: true, completion: nil)
}
```

Testing

Xcode Simulator **I/O** → **External Displays** → **CarPlay** (800×480 @2x): no **locked** iPhone, **Siri** or **audio**. **CarPlay Simulator** (Mac app, Additional Tools for Xcode) drives a **real iPhone** over a cable.

Traps

- ! Model built in the phone `SceneDelegate`: CarPlay may launch you with *no* phone scene.
- ! Locked phone: `.complete` files and `WhenUnlocked` keychain items fail — use `completeUntilFirstUserAuthentication / AfterFirstUnlock`.
- ! Hard-coded row counts: read the limits at runtime; they shrink while driving.
- ! “Custom CarPlay UI”: only the nav map. No video (bar parked video apps), games or free text.

Per category

- **Audio:** browse → `CPNowPlayingTemplate.shared`, a singleton fed by `MPNowPlayingInfoCenter`; its buttons and the steering wheel go through `MPRemoteCommandCenter` — the lock-screen plumbing.
- **Communication:** `SiriKit` (`INSendMessageIntent`, `INSearchForMessagesIntent`, `INStartCallIntent`) + `CallKit`; `CPContactTemplate`, `CPMessageListItem`. Siri dictates — no keyboard.
- **Navigation:** `CPMapTemplate` (buttons, trips, maneuvers via `CPNavigationSession`) over your map; `CPSearchTemplate`; `Dashboard` (`CPSupportsDashboardNavigationScene`) + instrument-cluster scenes.
- **EV / parking / food:** `CPPointOfInterestTemplate` (map + picker), `CPInformationTemplate` (items + ≤ 3 `CPTextButtons`).

New in iOS 26 · 26.4 · 27

- **26:** `.systemSmall` widgets + Live Activities (`.small`) show on the Dashboard, no CarPlay app needed; opt out: `.disfavoredLocations([.carPlay], for: [.systemSmall])`.
- **26.4:** voice conversational apps; food/driving-task depth 3.
- **27:** **video** apps — parked, in cars with video-in-car; AirPlay video; audio-only fallback (supportsVideoPlayback). `CPVoiceControlTemplate` for all.
- **CarPlay Ultra** (every car display): automaker feature; templates restyle.

Remember

“The entitlement picks the templates, templates are data, the car draws — only maps get pixels.”